

DSB6000 – Data Science Strategy & Leadership

NAPA Olympic Analytics Platform Consulting Competition

Graduate Data Science Consulting Capstone

Purpose and Use of This Overview

This document provides the high-level assignment orientation for the NAPA Olympic Analytics Platform Consulting Competition. Detailed submission requirements, evidence expectations, rubrics, and Canvas submission instructions are contained in the separate Phase 0 and Milestone assignment documents. When requirements differ, the specific supporting assignment document controls the submission for that phase or milestone.

1. Executive Scenario

The fictional North American Pickleball Association (NAPA) is preparing for the future inclusion of pickleball as an Olympic sport. To support this effort, NAPA has launched an analytics modernization initiative focused on athlete evaluation, Olympic roster selection, doubles partnership analysis, tournament simulation, and long-term athlete development.

Student teams will operate as competing analytics consulting firms responding to NAPA's request for a prototype analytics platform and evidence-based Olympic team selection recommendations. The work requires teams to connect business understanding, data engineering, governance, analytics, explainability, and executive communication into a coherent consulting response.

The assignment is not simply a ranking exercise. Consulting firms are expected to design and defend a data-driven approach capable of supporting ongoing Olympic roster selection, talent identification, and strategic decision-making under uncertainty.

2. Relationship to Supporting Assignment Documents

As the NAPA case has matured, detailed requirements have been moved into dedicated supporting documents for each delivery increment. This overview therefore summarizes the engagement model and points students to the appropriate detailed requirement documents rather than repeating every submission instruction in multiple places.

Supporting Document	Primary Purpose	Controls / Clarifies
Phase 0: Team Charter and Environment Readiness Requirements	Organizes the consulting team and confirms early environment readiness.	Team roster, role assignments, GitHub setup, Databricks readiness, working model, risks, and 10-point rubric.
Milestone 1: Foundational Readiness Requirements	Confirms the platform foundation using the 5K development dataset.	Raw landing, Bronze creation, catalog/medallion draft, row counts, schema validation, release discipline, and 20-point rubric.
Milestone 2: Engineering Readiness Requirements	Confirms the engineering solution using the 50K validation dataset.	Silver entities, quality rules, validation and reconciliation, governance documentation, early analytics, scaling risks, and 30-point rubric.
Milestone 3: Prototype Build / Final Submission and RFP Evaluation Committee Presentation Requirements	Serves as the final consulting submission and executive presentation package using the 250K production dataset.	Gold products, selection methodology, tournament candidate IDs, governance maturity, data quality confidence, executive deck, and 50-point rubric.

3. Assignment Objectives

This assignment simulates a graduate-level data science consulting engagement in which teams must design, build, analyze, and communicate a realistic analytics solution for a complex sports strategy problem. Students are not expected to build a production-ready enterprise platform. The objective is to demonstrate the ability to design, implement, analyze, govern, and defend a prototype analytics platform that supports Olympic athlete evaluation and decision-making.

By the end of the assignment, students should be able to demonstrate the following capabilities:

1. **Frame a business problem as a data science consulting opportunity.** Interpret NAPA's strategic goals and translate them into analytical questions, platform requirements, and measurable decision-support outcomes.
2. **Design a practical analytics architecture.** Describe how data ingestion, medallion architecture, validation, analytics, reporting, and future extensibility fit together.
3. **Apply data engineering and data quality practices.** Build repeatable workflows that manage realistic data conditions, increasing scale, validation requirements, and imperfect records.
4. **Use analytics, modeling, and simulation thoughtfully.** Evaluate player performance, doubles partnership effectiveness, regional competitiveness, development trends, uncertainty, and future potential.
5. **Explain analytical results in a defensible way.** Justify methods, assumptions, limitations, confidence levels, and recommendation logic for technical and executive stakeholders.
6. **Practice governance and responsible AI-assisted development.** Address reproducibility, lineage, data reliability, transparency, maintainability, and responsible use of AI tools.
7. **Operate as an effective consulting team.** Organize responsibilities, manage work across phases, use source control, document decisions, coordinate deliverables, and collaborate professionally.
8. **Communicate with executive stakeholders.** Translate complex technical work into concise, business-relevant recommendations for the NAPA RFP Evaluation Committee.

4. Business Problem

NAPA leadership requires a repeatable, data-driven approach for evaluating athletes, identifying high-performing doubles partnerships, selecting Olympic rosters, and developing future talent pipelines for both Team USA and Team Canada.

Consulting firms must develop analytical methodologies capable of supporting decision-making in an environment where player performance, partnerships, competitive conditions, and data quality may vary over time. Solutions should incorporate data engineering, analytics, explainability, governance, and executive decision support.

The challenge is not simply to identify the strongest athletes today. Teams must provide NAPA leadership with a defensible framework for evaluating current performance, future potential, uncertainty, and risk when making Olympic roster decisions.

5. Consulting Competition Structure

The class will be divided into six competing analytics consulting firms. Each consulting firm represents an independent sports analytics and technology consultancy competing for the opportunity to become the official Olympic analytics partner for NAPA.

Teams should approach the engagement as a realistic enterprise analytics procurement and consulting selection process. The strongest submissions will demonstrate not only technical execution, but also professional judgment, governance maturity, analytical rigor, and executive communication.

6. Team Organization and Roles

Each consulting firm should operate as a professional analytics consulting organization with responsibilities distributed across technical, analytical, governance, and business communication functions. Detailed role expectations, required evidence, and submission format are provided in the Phase 0: Team Charter and Environment Readiness Requirements document.

Roles are intended to clarify accountability and organize the work. They are not intended to restrict participation or create hard boundaries. All team members remain jointly accountable for the final quality of the consulting submission.

7. Dataset Release Strategy

Consulting firms will receive historical pickleball data representing competitive activity across the United States and Canada. The dataset will be released in three increments to support progressive development, validation, and final production-scale analysis.

Dataset Increment	Primary Purpose	Expected Use
5K development dataset	Small-scale development and platform foundation.	Used for Milestone 1 to confirm repository discipline, configuration approach, source landing, Bronze creation, row counts, and schema validation.
50K validation dataset	Engineering scale-up and quality validation.	Used for Milestone 2 to confirm repeatable execution, Silver-layer entities, data quality rules, validation evidence, governance documentation, and early analytics.
250K production dataset	Final prototype build and recommendation evidence.	Used for Milestone 3 to complete the prototype analytics platform, Gold products, Olympic selection methodology, tournament candidate selection, and executive presentation.

A central technical requirement is that teams should be able to move across the 5K, 50K, and 250K datasets using configuration changes rather than structural code rewrites. Hard-coded paths, account-specific assumptions, and manual rewrites between dataset sizes should be treated as engineering defects.

8. Delivery Roadmap

The work is organized into a sequence of Canvas-managed submissions. The overview below is intentionally concise; teams must consult the supporting assignment documents for the detailed requirements, evidence expectations, and rubrics for each submission.

Canvas Delivery	Dataset	Primary Focus	Detailed Requirements
Pre-Onboarding Readiness	None	Individual skills assessment and self-paced preparation.	Tracked separately by the instructor.
Phase 0: Team Charter and Environment Readiness	None / setup	Team formation, role ownership, GitHub setup, Databricks readiness, and project understanding.	Phase 0 supporting assignment document.
Milestone 1: Foundational Readiness	5K	Raw landing, Bronze-layer creation, catalog/medallion draft, and basic validation evidence.	Milestone 1 supporting assignment document.
Milestone 2: Engineering Readiness	50K	Silver-layer engineering, quality rules, validation and reconciliation, governance documentation, early analytics, and scaling observations.	Milestone 2 supporting assignment document.
Milestone 3: Prototype Build / Final Submission	250K	Gold-layer analytics, final recommendations, tournament candidates, governance maturity, data quality confidence, GitHub evidence, and final RFP Evaluation Committee presentation.	Milestone 3 supporting assignment document.

9. Olympic Team Construction Requirements

A primary objective of the engagement is to develop a data-driven framework for identifying both current Olympic-caliber doubles teams and future elite talent. Each consulting firm must recommend Olympic roster candidates for both Team USA and Team Canada.

At the final submission stage, teams will recommend teams and alternates for men's doubles, women's doubles, and mixed doubles, supported by analytical evidence and a clearly defined evaluation methodology. Teams should consider factors such as player performance, partnership effectiveness, consistency, strength of competition, development trends, uncertainty, and projected future success.

Final roster recommendations, tournament candidate IDs, evidence expectations, and presentation requirements are controlled by the Milestone 3 supporting assignment document.

10. Expected Deliverable Categories

The full engagement requires a blend of technical, analytical, governance, and executive deliverables. The categories below summarize the expected body of work at a high level; detailed evidence and scoring expectations are defined in the phase and milestone documents.

Category	High-Level Expected Work
Technical and Engineering	Repository discipline, configuration-driven execution, medallion architecture, raw-to-Bronze ingestion, Silver business entities, Gold data products, validation evidence, and reproducibility.
Data Governance and Quality	Catalog design, data dictionary, lineage documentation, quality rule catalog, quality reports, ownership assumptions, classification, access-control considerations, and analytical confidence discussion.
Analytics and Decision Support	Exploratory findings, ranking and scoring methodology, player and partnership evaluation, Olympic roster recommendations, development pipeline insights, uncertainty treatment, and tournament candidate selection.
Business and Executive Communication	Professional executive presentation, strategic findings, risks, assumptions, limitations, implementation roadmap, and clear recommendations for NAPA leadership.

11. Final Executive Presentation

The final executive presentation is consolidated into Milestone 3: Prototype Build / Final Submission. Each consulting firm will deliver a formal presentation to the NAPA RFP Evaluation Committee that explains the proposed analytics platform, governance approach, analytical evidence, selection methodology, recommendations, risks, and implementation roadmap.

The presentation should be structured as a professional consulting readout for both technical leaders and non-technical executives. It should emphasize business value, decision confidence, and strategic implications rather than technical implementation details alone. Detailed presentation content, submission format, and rubric expectations are provided in the Milestone 3 supporting assignment document.

12. Tournament Simulation and Championship

Using team submissions, we will execute a tournament simulation using Monte Carlo analyses, including a final "live" run following the presentations, in the final class of the term.

The tournament is intended to add realism, comparison, and energy to the final class experience. Teams will submit eligible team IDs selected from the NAPA dataset. The tournament outcome may be recognized separately in class, but it will not determine the assignment grade. Student evaluation will focus on the quality of the

analytical process, supporting evidence, technical implementation, governance discipline, and executive recommendations.

13. Live Committee Evaluation

The class may serve as the NAPA RFP Evaluation Committee and provide live feedback on final presentations. These evaluations are intended to support discussion, realism, and peer learning. The instructor remains responsible for grading each assignment using the published milestone rubrics.

14. Grading and Evaluation Model

Canvas will be used as the official control point for submissions, due dates, rubrics, feedback, and grade tracking. The assignment is assessed through progressive team submissions: Phase 0, Milestone 1, Milestone 2, and Milestone 3. Each milestone has its own supporting assignment document and rubric.

Grades will reward the ability to combine sound architecture, reliable engineering, governance maturity, data quality discipline, analytical rigor, responsible AI-assisted development, and executive clarity. Tournament outcomes and live committee feedback are not substitutes for the instructor's rubric-based grading.

15. Final Goal

The objective is to develop and defend an analytically rigorous NAPA analytics platform prototype and an effective Olympic roster strategy recommendation under uncertainty.

Strong submissions will combine technical competence, strategic reasoning, simulation awareness, explainability, governance discipline, and compelling executive storytelling.

What Matters Most for This Case Study

For the purposes of this project, students should focus on the following questions:

- Which athletes are most likely to succeed at the Olympic level?
- Which doubles partnerships perform best together?
- Which athletes are improving most rapidly?
- Which partnerships have the greatest future potential?
- How confident can NAPA be in roster recommendations?
- What analytical evidence supports those recommendations?

The goal of this engagement is not to learn the rules of pickleball. The goal is to use data, analytics, governance, and strategic reasoning to support high-quality athlete selection and development decisions, in conjunction with demonstration of the teams analytics approach and technical capabilities, intended to provide NAPA with sufficient information to select a long-term partner.

Appendix A - Pickleball Overview

This appendix provides a brief overview of pickleball concepts relevant to the NAPA Olympic Analytics Case Study. Students are not expected to become experts in the sport; however, a basic understanding of competitive pickleball will help inform analytical decisions and roster recommendations. A thorough understanding of all pickleball rules and regulations is NOT required for this assignment.

What is Pickleball?

Pickleball is a paddle sport that combines elements of tennis, badminton, and table tennis. It is played at recreational, amateur, and professional levels and has become one of the fastest-growing sports in North America.

The sport can be played as either singles or doubles. While singles competition exists, elite-level competition is heavily focused on doubles play, making partnership selection a critical component of this case study.

Doubles Competition

Doubles teams consist of two players competing together against another two-player team. Competitive pickleball includes three primary doubles categories:

- Men's Doubles
- Women's Doubles
- Mixed Doubles

Because this case study focuses on Olympic team selection, consulting firms will evaluate athletes and partnerships within all three categories.

Match and Game Structure

Pickleball matches are composed of one or more games. For purposes of this case study, it is sufficient to understand that games are typically played to 11 points, a team must win by at least 2 points, and strong teams generally demonstrate the ability to consistently win games and matches against quality competition.

The datasets provided by NAPA will contain game-level and match-level results that can be used to evaluate player and partnership performance.

Why Doubles Partnerships Matter

Strong individual athletes do not always form strong doubles teams. Partnership success may be influenced by communication, consistency, chemistry, experience playing together, pressure performance, and complementary strengths and weaknesses. NAPA is therefore interested in identifying both strong individual athletes and strong combinations of athletes.

Player Ratings and Rankings

Competitive pickleball commonly uses rating systems to estimate player skill levels. Ratings and rankings can help compare athletes, seed tournaments, evaluate competitive progression, and estimate future performance. However, ratings alone may not tell the complete story. A key objective of this engagement is to determine whether additional analytical methods can improve athlete and partnership evaluation.

